

An autonomous feedback protocol

Responding to temperature and potential changes in energy converters

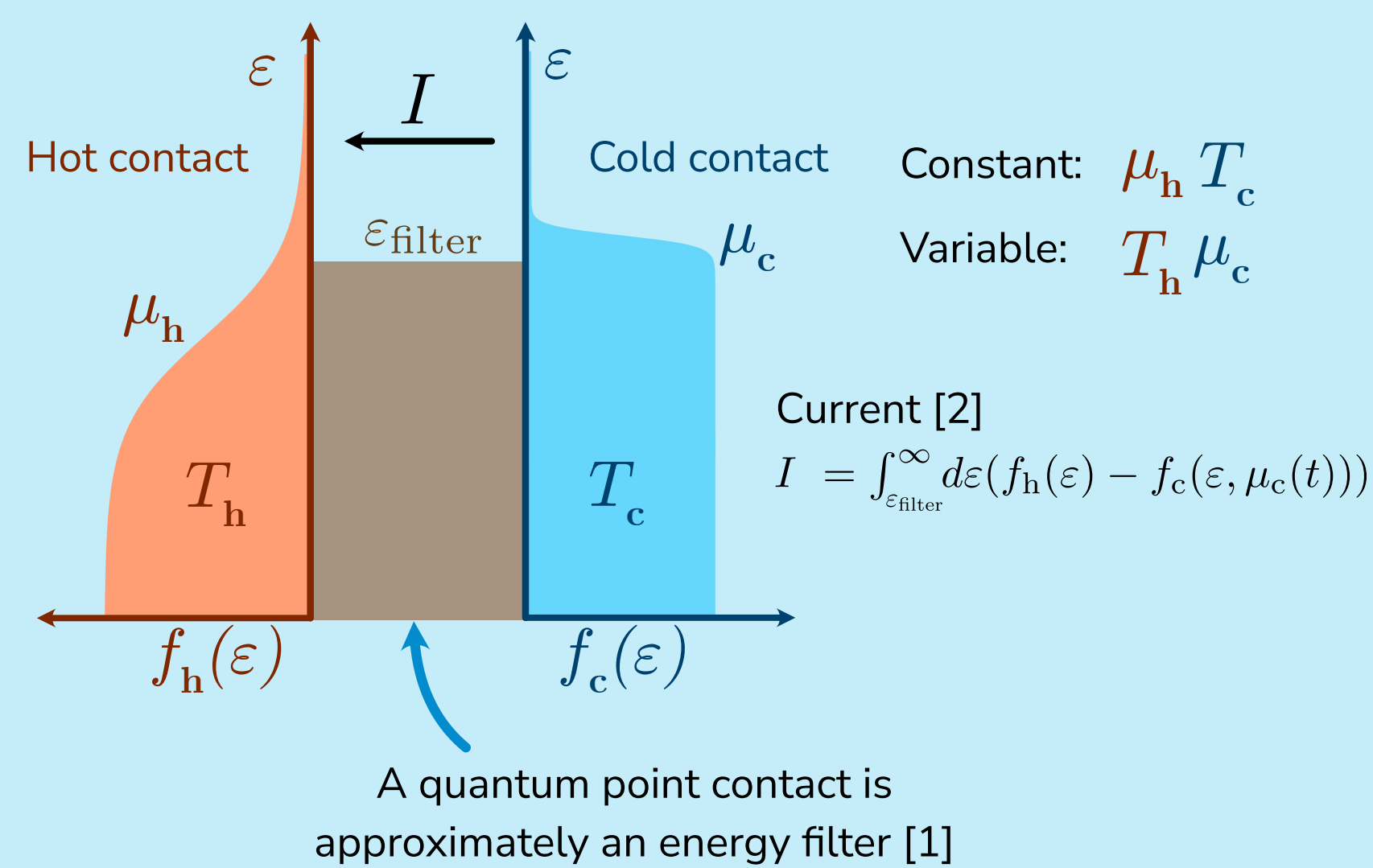
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Introduction

Is it possible for a quantum heat engine to adapt to its available resources *without external control*? We propose an autonomous feedback mechanism that optimizes the average performance of a nanoscale thermoelectric device. In our situation, external conditions induce a varying or unknown incoming temperature which provides the heat the device converts to electric power. But a device designed to work optimally at one temperature will have a lower effect at another, since the conductor typically has static thermoelectric properties. Our feedback mechanism autonomously alters those properties, enabling higher average outputs. Here we show how the device is designed, along with the principles behind it. We also discuss timescales for different processes, which informs the valid regimes. Finally, we demonstrate that our device outperforms static devices when temperature is known only on a probabilistic level.

Steady-state engine

Heat converts to work when electrons flow from **low** to **high** electrochemical potential.



Timescales

For our model to hold we need the timescale separations:

Thermalization $\ll$ Traversal $\ll$ Charging, dot dynamics

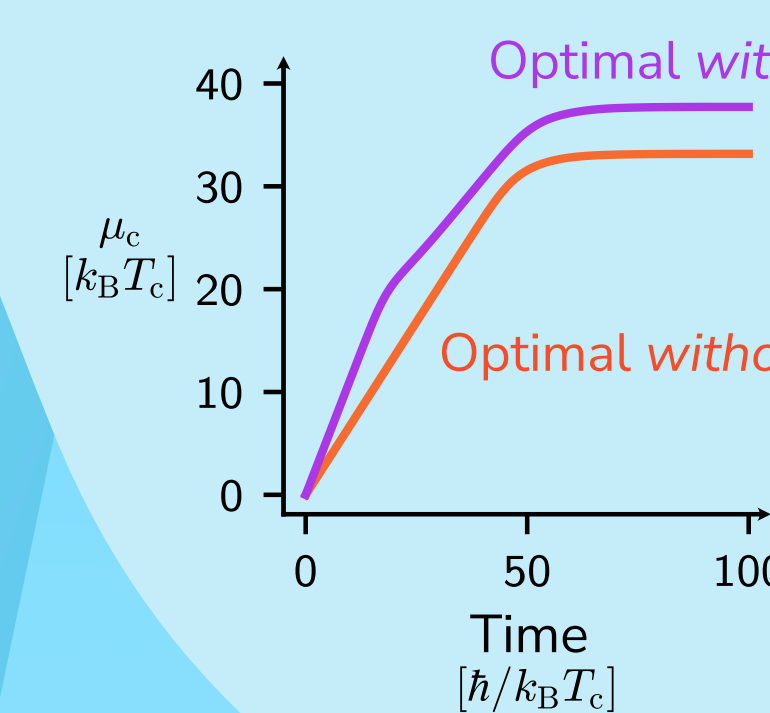
Distributions always thermalized Electrons arriving do not outpace thermalization Feedback mechanism always sees a steady-state device

Charging, the increase in potential due to current flow, and dot dynamics, the time between jumps in and out of the quantum dot, can also be separated. That leads to two distinct model setups.

Charging $\gg$ Dot dynamics

Battery setup

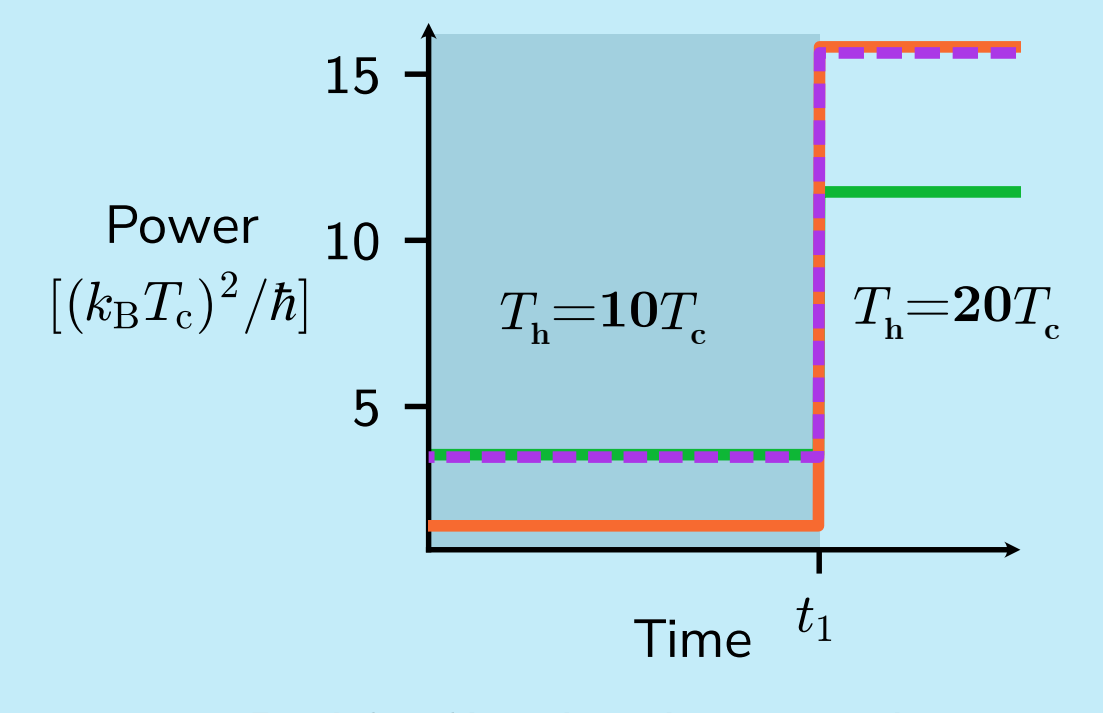
- Potential changes due to accumulation of charge
- The quantum dot fills *because* the potential increases
- Feedback increases the charge build-up:



Charging $\ll$ Dot dynamics

Load setup

- Potential changes due to Ohm's law in external resistance
- The quantum dot filling *causes* potential to increase
- Feedback increases the power output:



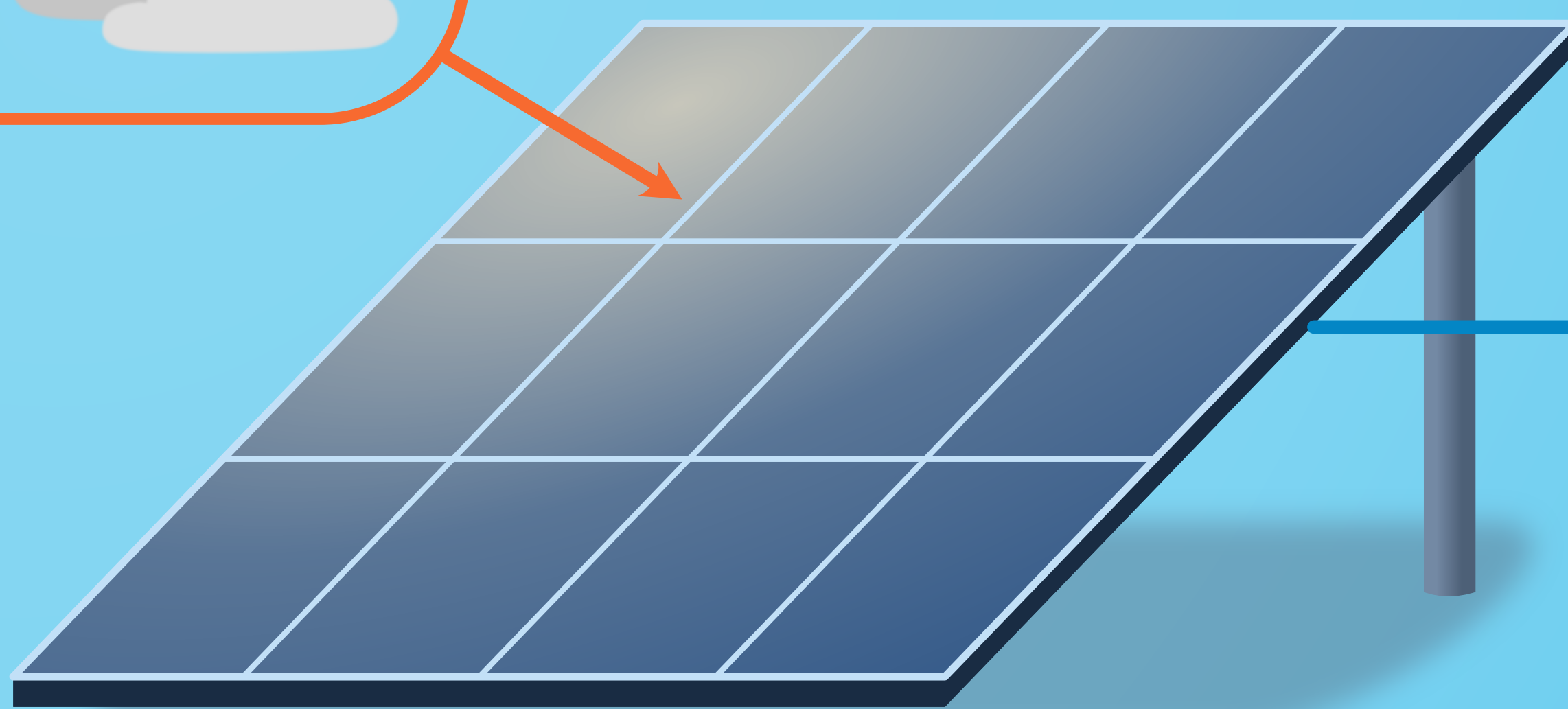
Red for filter height optimal at $T_h = 20T_c$, green at $T_h = 10T_c$
Purple with feedback is optimal at both temperatures

The Inspiration

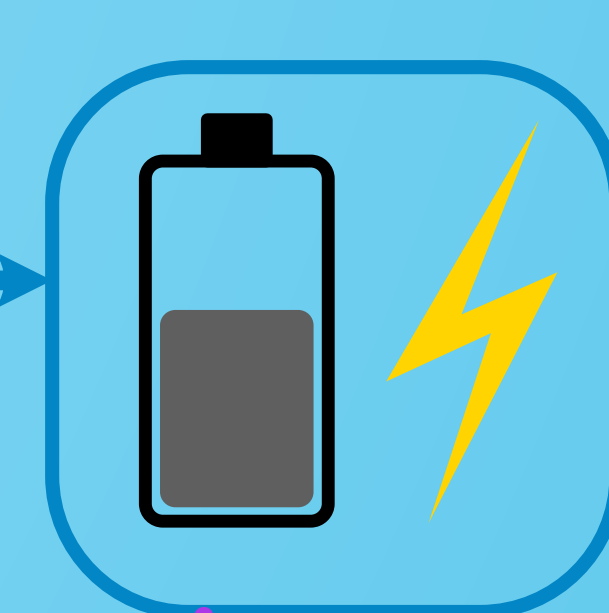
Heat source



Conductor

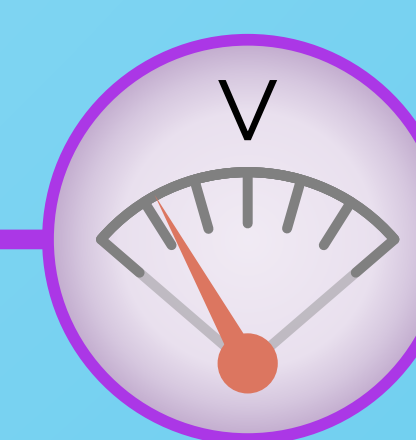


Cold contact



Autonomous feedback

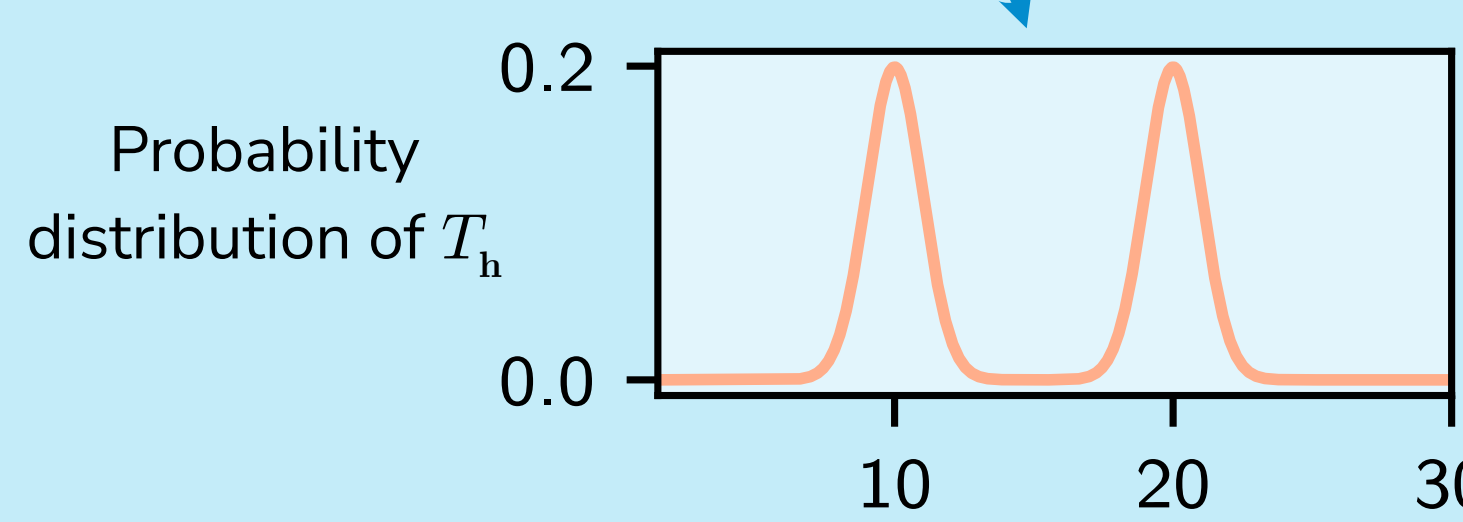
Measure potential



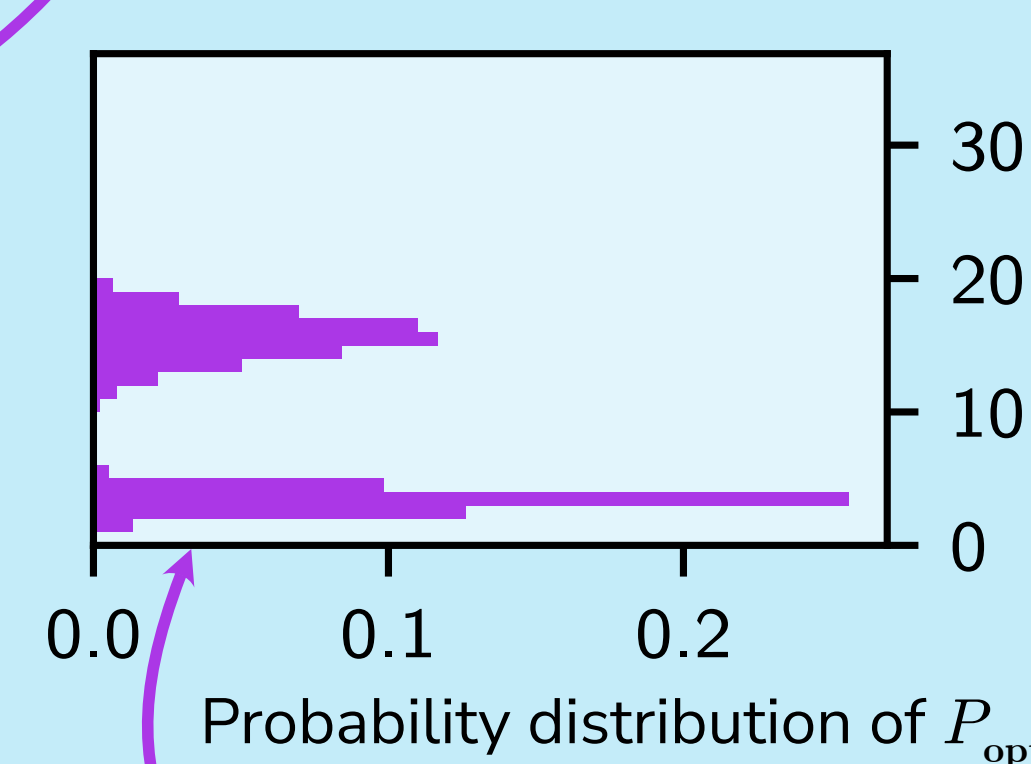
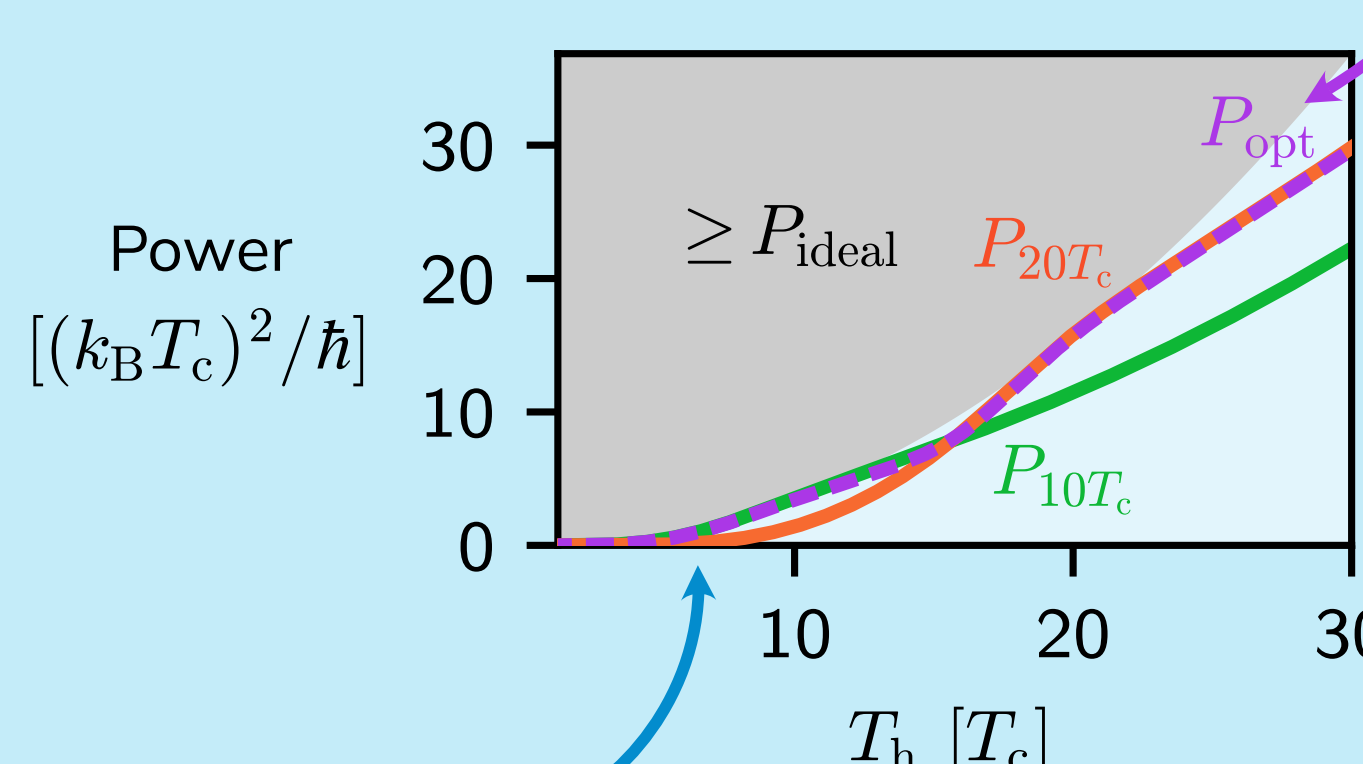
Temperature distribution

For the load setup, we assume the incoming temperature follows a probability distribution. The goal is to maximize the average power output, informed by the temperature distribution

Pick a bimodal distribution around two operating temperatures, for instance



Optimized feedback gives generally higher power for different temperatures



Red for filter height optimal at $T_h = 20T_c$, green at $T_h = 10T_c$

Used to find the average power value. It is higher for feedback than nonfeedback:

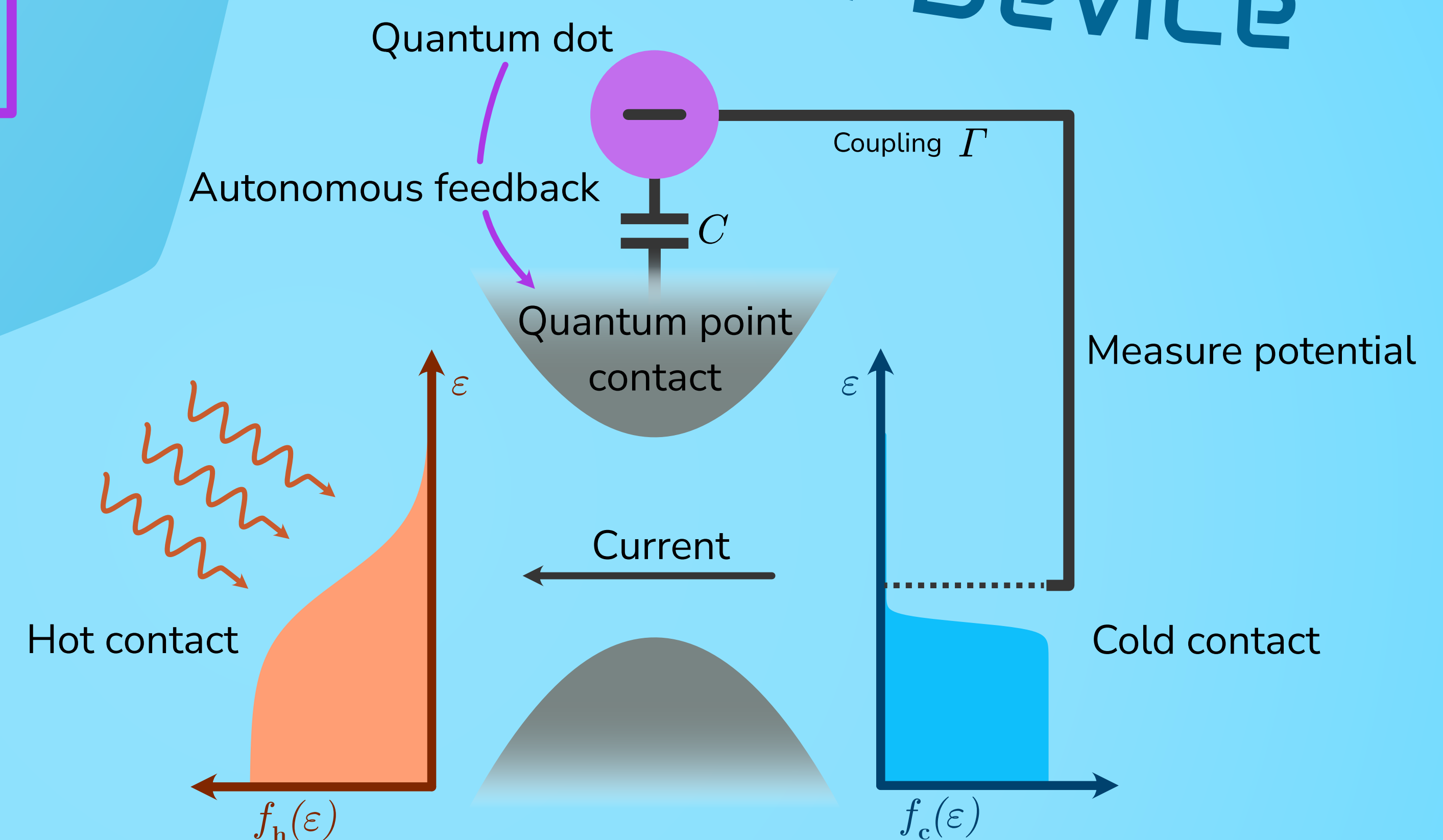
$$\bar{P}_{10T_c} = 7.51$$

$$\bar{P}_{20T_c} = 8.62$$

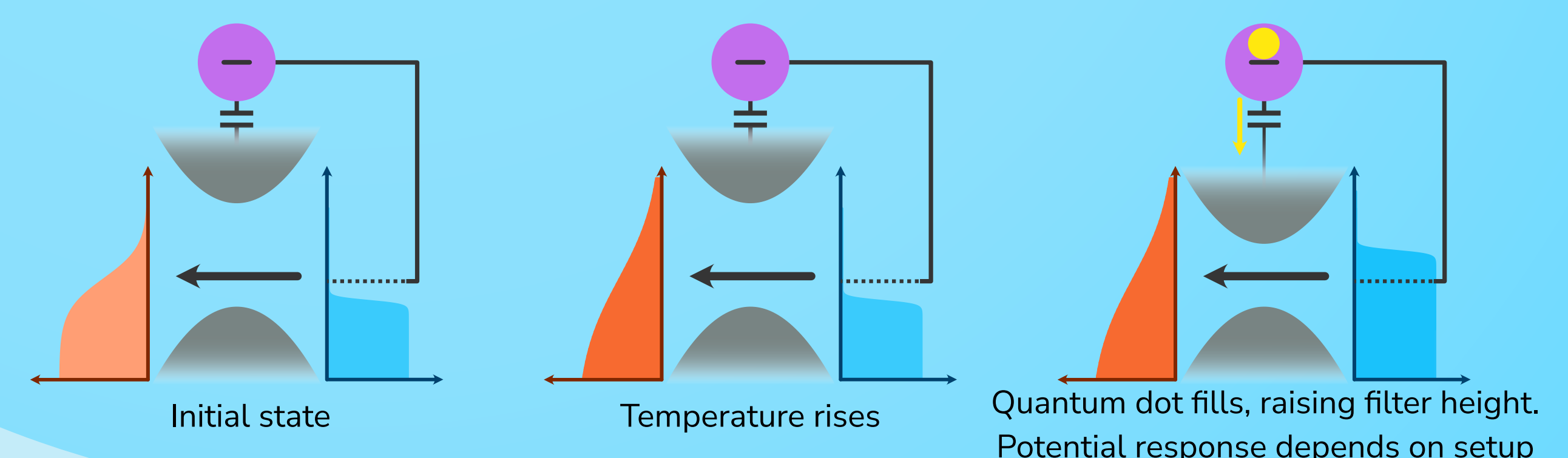
$$\bar{P}_{\text{opt}} = 9.53$$

$$[(k_B T_c)^2 / h]$$

The Device



Feedback mechanism



Conclusion

Autonomous feedback increases the performance of a nanoscale, thermoelectric device by adapting to changes in temperature and potential.

- Experimentally realizable device, already used to detect single-electron events in a quantum dot, e.g. [3][4].
- We propose to *exploit* the current through the quantum point contact instead.
- Output, power or charge build-up, increases significantly with feedback.
- Useful when external conditions, affecting the incoming temperature, are variable or unknown: **the device adapts**.
- Future studies? Experimental implementation, explicitly time-dependent temperature, fluctuations, multi-level or continuous charge in detector.

References

- [1] M. Büttiker, Phys. Rev. B 41, 7906-7909 (1990).
- [2] M. V. Moskalets, Scattering Matrix Approach to Non-Stationary Quantum Transport
- [3] W. Lu, Z. Ji, L. Pfeiffer, K. W. West, and A. J. Rimberg, Nature 423, 422-425 (2003).
- [4] S. Gustavsson, R. Leturcq, B. Simović, R. Schleser, T. Ihn, P. Studerus, K. Ensslin, D. C. Driscoll, and A. C. Gossard, Phys. Rev. Lett. 96, 076605 (2006).